

ECONOMICS

FOURTH EDITION

14(29)

Money, Banking, and the Federal Reserve System

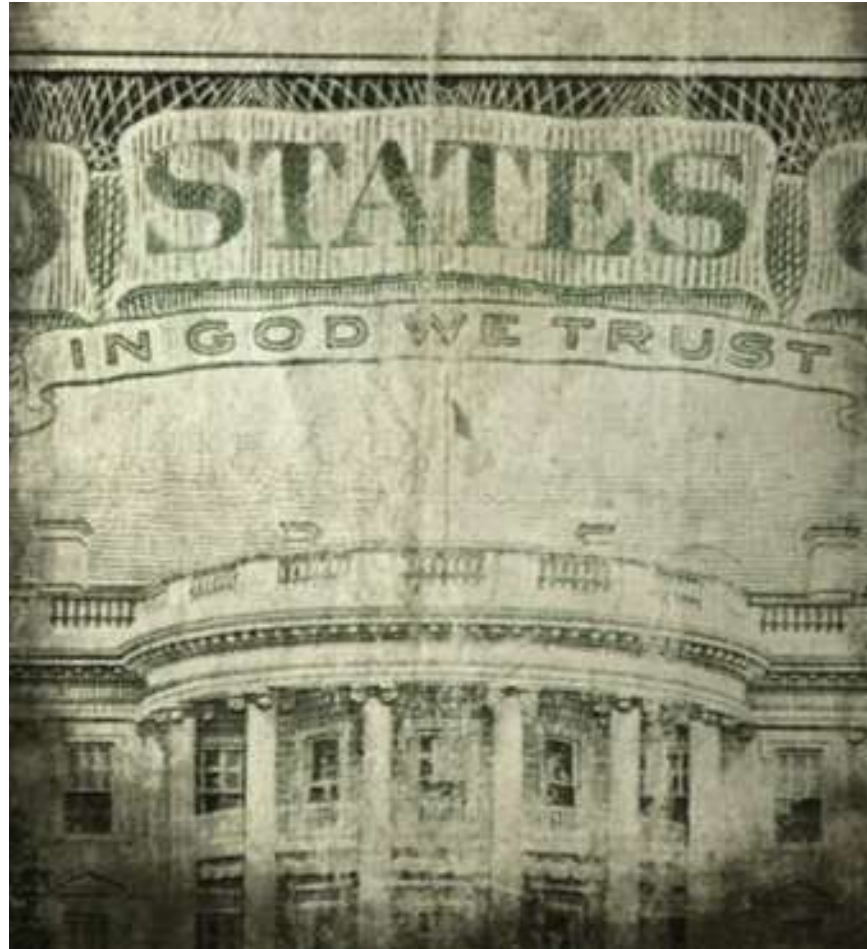
▶ What you will learn in this chapter



- The various roles money plays and the many forms it takes in the economy
- How the actions of private banks and the Federal Reserve determine the money supply
- How the Federal Reserve uses open-market operations to change the monetary base

MONEY: THE ESSENTIAL CHANNEL BETWEEN MARKETS

Money plays unique and very important roles in modern economies.



THE MEANING OF MONEY

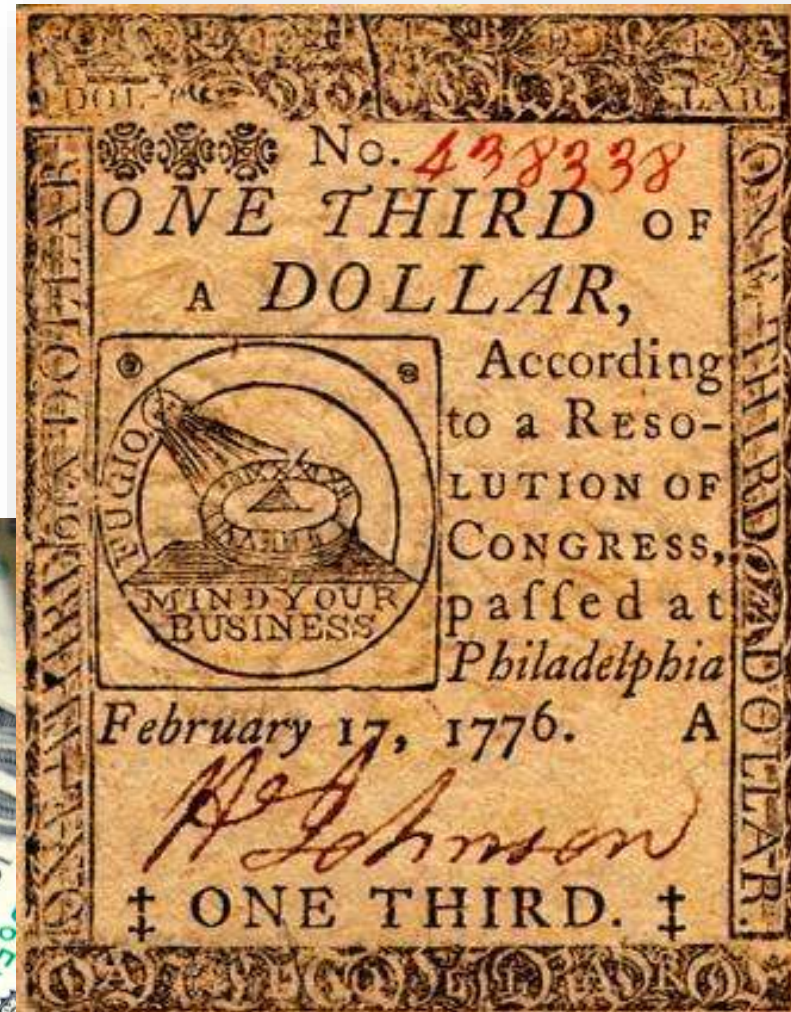
Money: any asset that can easily be used to purchase goods and services.



In 1933, after many bank failures, some U.S. towns turned to homemade currency, like this clamshell scrip used in Pismo Beach, California.

THE MEANING OF MONEY

Currency in circulation:
cash held by the public.



Early U.S. currency:
continental
third of a dollar

THE MEANING OF MONEY

Checkable bank deposits: bank accounts on which people can write checks.



THE MEANING OF MONEY



The **money supply** is the total value of financial assets in the economy that are considered money.

THE ROLES OF MONEY

Money must function as:

- *a medium of exchange.*
- *a store of value.*
- *a unit of account.*



THE FUNCTIONS OF MONEY

Medium of exchange: Something people accept as payment for goods and services



(an asset that individuals acquire **for the purpose of trading** rather than for their own consumption).

THE FUNCTIONS OF MONEY

Store of value: Money is a means of holding purchasing power over time.

It enables people to save the money they earn today and use it to buy the goods and services they want tomorrow.



THE FUNCTIONS OF MONEY



Unit of account: Money provides a yardstick for measuring and comparing the values of a wide variety of goods and services.

TYPES OF MONEY

For thousands of years, societies have used **commodity money**.

Commodity money: a good used as a medium of exchange that **has intrinsic value in other uses**.



Some prisons use cigarettes as currency.



The island of Yap's currency

TYPES OF MONEY

To make life simpler (and safer), societies began using commodity-backed money.

Commodity-backed money: a medium of exchange with **no intrinsic value** whose ultimate value is guaranteed by a promise that **it can be converted into valuable goods.**



ANOTHER EXAMPLE OF COMMODITY-BACKED MONEY



TYPES OF MONEY

Most modern economies use fiat money.

Fiat money: money whose value derives entirely from its official status as a means of payment.



LEARN BY DOING: DISCUSS

**With a partner,
discuss:
How would the
economy be
different if we used
only **local
currency?****



MEASURING THE MONEY SUPPLY

Monetary aggregate: an overall measure of the money supply.

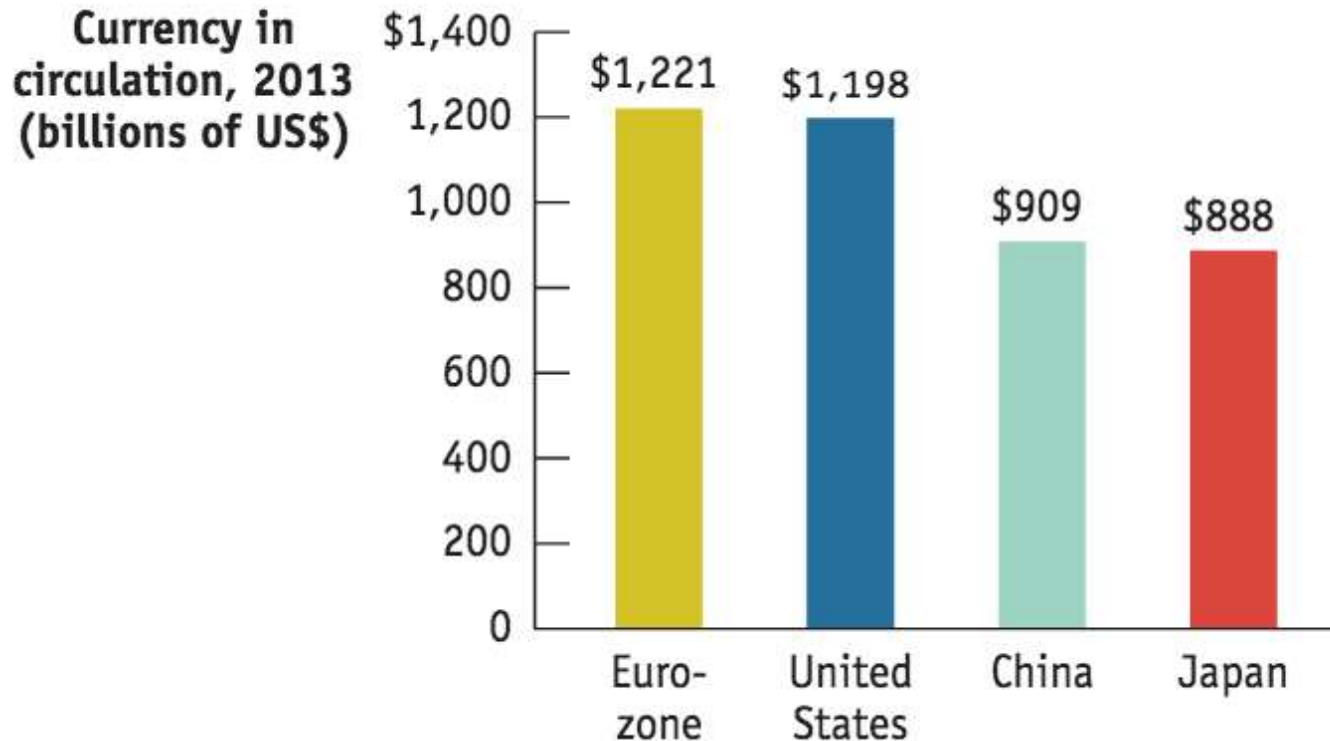
M1 and M2: two definitions of the money supply.



THE BIG MONIES



The U.S. Dollar is still the currency most likely to be accepted for payment around the world... but is not the biggest money.

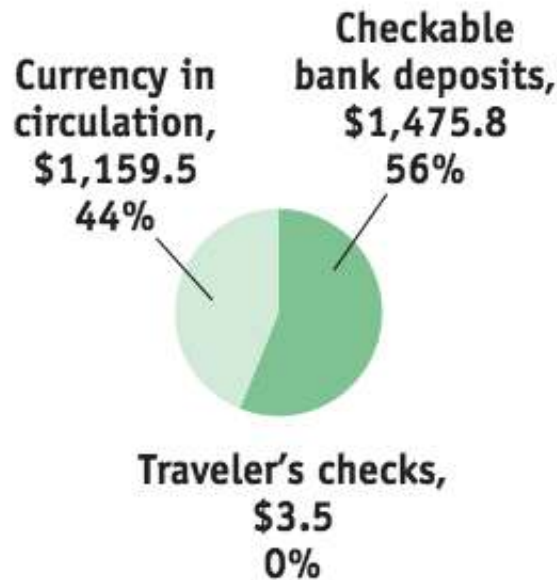


MONETARY AGGREGATES, 2013

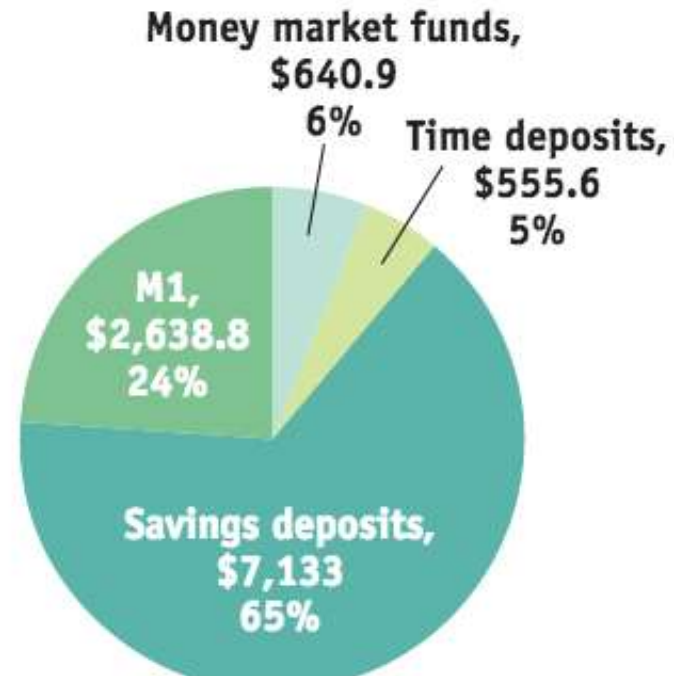
M1 includes only the most liquid forms of money.

M2 includes *near-moneys*: financial assets that can't be directly used as a medium of exchange but can readily be converted into cash or checkable bank deposits.

(a) M1 = \$2,638.8
(billions of dollars)



(b) M2 = \$10,968.3
(billions of dollars)



Source: Federal Reserve Bank of St. Louis.

LEARN BY DOING: PRACTICE QUESTION

Which of the following assets would you classify as being most liquid?

- a) demand deposits**
- b) small-time deposits**
- c) a home**
- d) gold bullion**

THE MONETARY ROLE OF BANKS



Q: Only half of $M1$ is currency; where does the rest of the money supply come from?

A: Bank deposits make up half of $M1$ and most of $M2$.

How does the banking system create money?

WHAT BANKS DO

Banks are financial intermediaries that use **liquid assets** (in the form of bank deposits) to finance the **illiquid investments** of borrowers.



THE MONETARY ROLE OF BANKS

T-account: a tool for analyzing a business's financial position by showing the business's assets and liabilities.

Assets		Liabilities	
Loans	\$1,200,000	Deposits	\$1,000,000
Reserves	\$100,000		

Bank reserves: the currency that banks hold in their vaults plus their deposits at the Federal Reserve.

The reserve ratio: the fraction of bank deposits that a bank holds as reserves
 $(\$100,000 / \$1,000,000) = 10\%$.

HOW BANKS CREATE MONEY

The primary purpose of financial markets and institutions **is to act as a bridge between lenders and borrowers.**

By accepting deposits and making loans, banks and other financial institutions are also able to create money.



MONEY CREATION

Financial institutions
operate as part of a
*fractional reserve banking
system.*

When you deposit money in
a bank account, the bank is
required to hold a part of it in
its vault as cash

*(or else in an account with
the regional Federal
Reserve Bank).*



*This system's
leverage makes
some nervous.*

THE PROBLEM OF BANK RUNS

**Banks hold only a fraction of deposits on reserve.
(The rest they use to make loans.)**

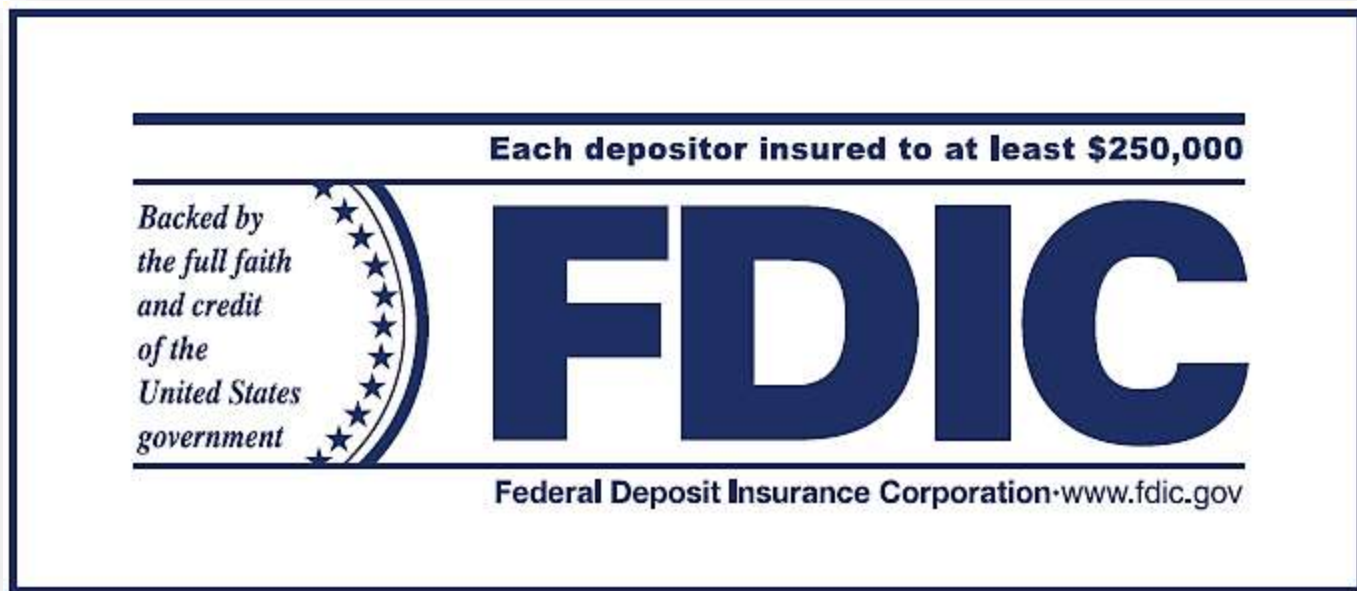
From time to time people get worried about the bank losing their money.

Bank run: a phenomenon in which many of a bank's depositors try to withdraw their funds because they fear a bank failure.

Historically, runs have often proved contagious.

BANK REGULATION

- 1. *Deposit insurance:*** a guarantee that a bank's depositors will be paid even if the bank can't come up with the funds. (The FDIC currently guarantees the first \$250,000 of each account.)



BANK REGULATION

2. *Capital requirements:* requirement that the owners of banks hold substantially more assets than the value of bank deposits.

Deposit insurance creates a well-known incentive problem: Banks can take more risks, since they are insured.

To help motivate safe behavior, banks' capital is required to equal to 7% or more of their assets.

BANK REGULATION

- 3. *Reserve requirements:*** rules set by the Federal Reserve that determine the minimum reserve ratio for a bank.

*For example, in the **United States**, the minimum **reserve ratio for checkable bank deposits** is 10%.*

- 4. *The discount window:*** an arrangement in which the Federal Reserve stands ready to lend money to banks in trouble.

HOW BANKS CREATE MONEY

Silas keeps a shoebox full of cash under his bed. Deciding to enter the twenty-first century, **he deposits this cash at the bank.** *What's the effect of his \$1,000?*

(a) Initial Effect Before Bank Makes a New Loan

Assets		Liabilities	
Loans	No change	Checkable deposits	+\$1,000
Reserves	+\$1,000		

(b) Effect When Bank Makes a New Loan

Assets		Liabilities
Loans	+\$900	No change
Reserves	-\$900	

Banks make their profit from loans, so eventually they will make new loans with 90% of Silas's money. (The rest they must hold in reserve.)

HOW BANKS CREATE MONEY

The first bank lends \$900 to Maya, who pays the money to Anne, who deposits it at her bank—and the cycle starts all over. Sound familiar?

	Currency in circulation	Checkable bank deposits	Money supply
First stage: Silas keeps his cash under his bed.	\$1,000	\$0	\$1,000
Second stage: Silas deposits cash in First Street Bank, which lends out \$900 to Maya, who then pays it to Anne Acme.	900	1,000	1,900
Third stage: Anne Acme deposits \$900 in Second Street Bank, which lends out \$810 to another borrower.	810	1,900	2,710

RESERVES, BANK DEPOSITS, AND THE MONEY MULTIPLIER

Excess reserves: a bank's reserves over and above its required reserves.

We'll assume banks don't want to hold excess reserves.

Increase in bank deposits from \$1,000 in excess reserves =

$$\$1,000 + [\$1,000 \times (1 - rr)] + [\$1,000 \times (1 - rr)^2] + [\$1,000 \times (1 - rr)^3] + \dots$$

This can be simplified to $\$1,000/rr$.

LEARN BY DOING: PRACTICE QUESTION

Assets		Liabilities	
Required reserves	\$100	Deposits	\$1,000
Loans	\$400		
Treasury bills	\$800		

Given this T-account and assuming the bank holds no excess reserves, what is the required reserve ratio?

- a) 10%**
- b) 40%**
- c) 80%**
- d) 1%**

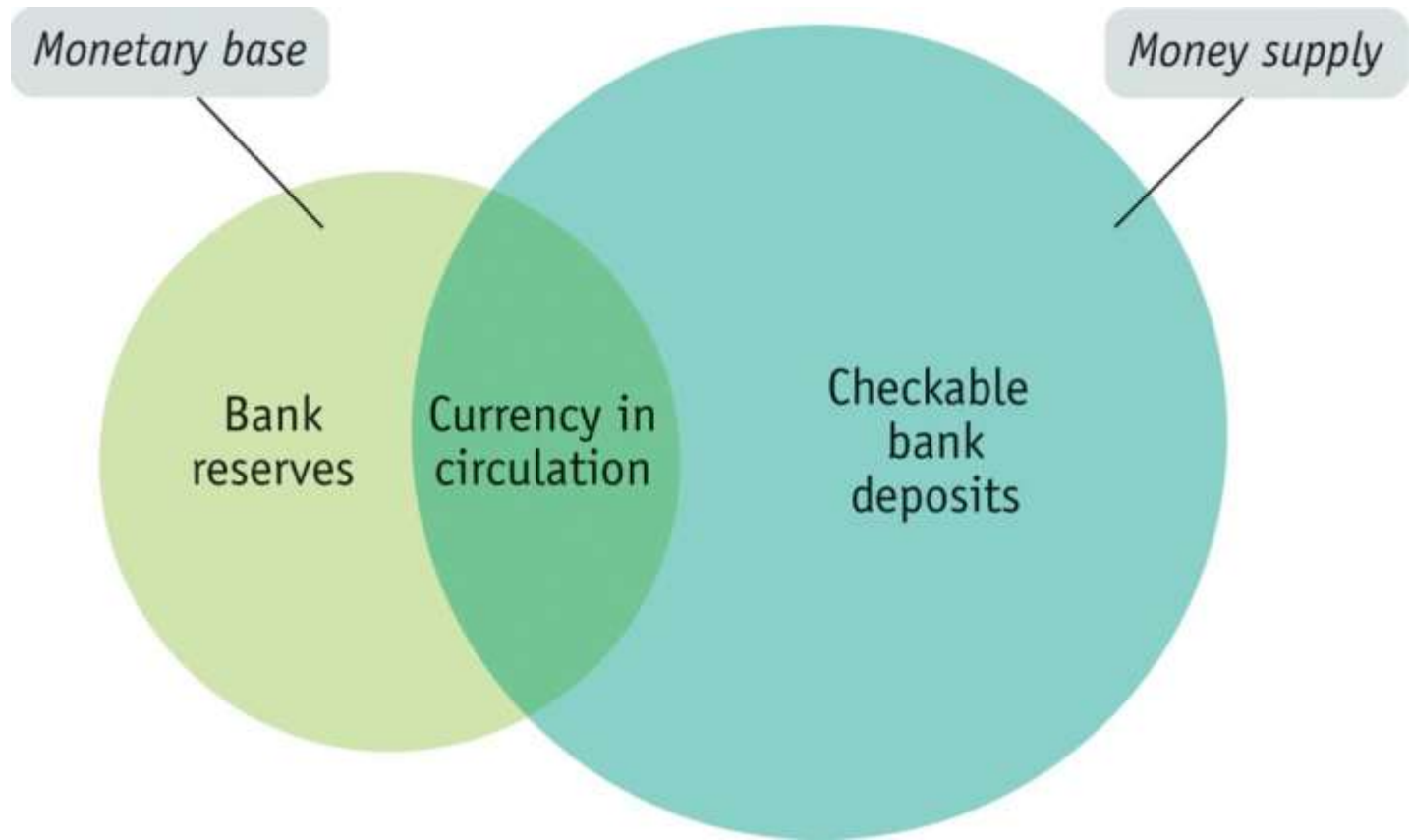
LEARN BY DOING: PRACTICE QUESTION

Suppose the required reserve ratio initially is 10% of bank deposits and is increased by the Fed to 20% of bank deposits. Holding everything else constant, this will:

- a) reduce the size of the money multiplier.**
- b) cause the banking system to contract the level of bank deposits in the banking system.**
- c) change the value of the money multiplier from 10 to 5.**
- d) Answers (a), (b), and (c) are all correct.**

THE MONEY MULTIPLIER IN REALITY

The **monetary base** is the sum of currency in circulation and bank reserves.



THE MONEY MULTIPLIER IN REALITY

In reality, the determination of the **money supply is more complicated** than our simple model suggests:

*It depends not only on the **ratio of reserves to bank deposits** but also on the **fraction of the money supply that individuals choose to hold in the form of currency.***



THE MONEY MULTIPLIER IN REALITY

The ***money multiplier*** is the ratio of the money supply to the monetary base.

In normal times, the U.S. money multiplier for M1 is between 1.5 and 3.0.

Why isn't it $1/0.1 = 10$?

People hold significant amounts of cash, which reduces bank deposits.





MULTIPLYING MONEY DOWN

If Silas can increase the money supply by pulling cash out from under his bed and injecting into the banking system, the reverse can happen anytime people carry cash.

Example: The Bank Runs of 1929-1933

	Currency in circulation	Checkable bank deposits	M1
	(billions of dollars)		
1929	\$3.90	\$22.74	\$26.64
1933	5.09	14.82	19.91
Percent change	+31%	−35%	−25%

Source: U.S. Census Bureau (1975), *Historical Statistics of the United States*.

THE FEDERAL RESERVE SYSTEM

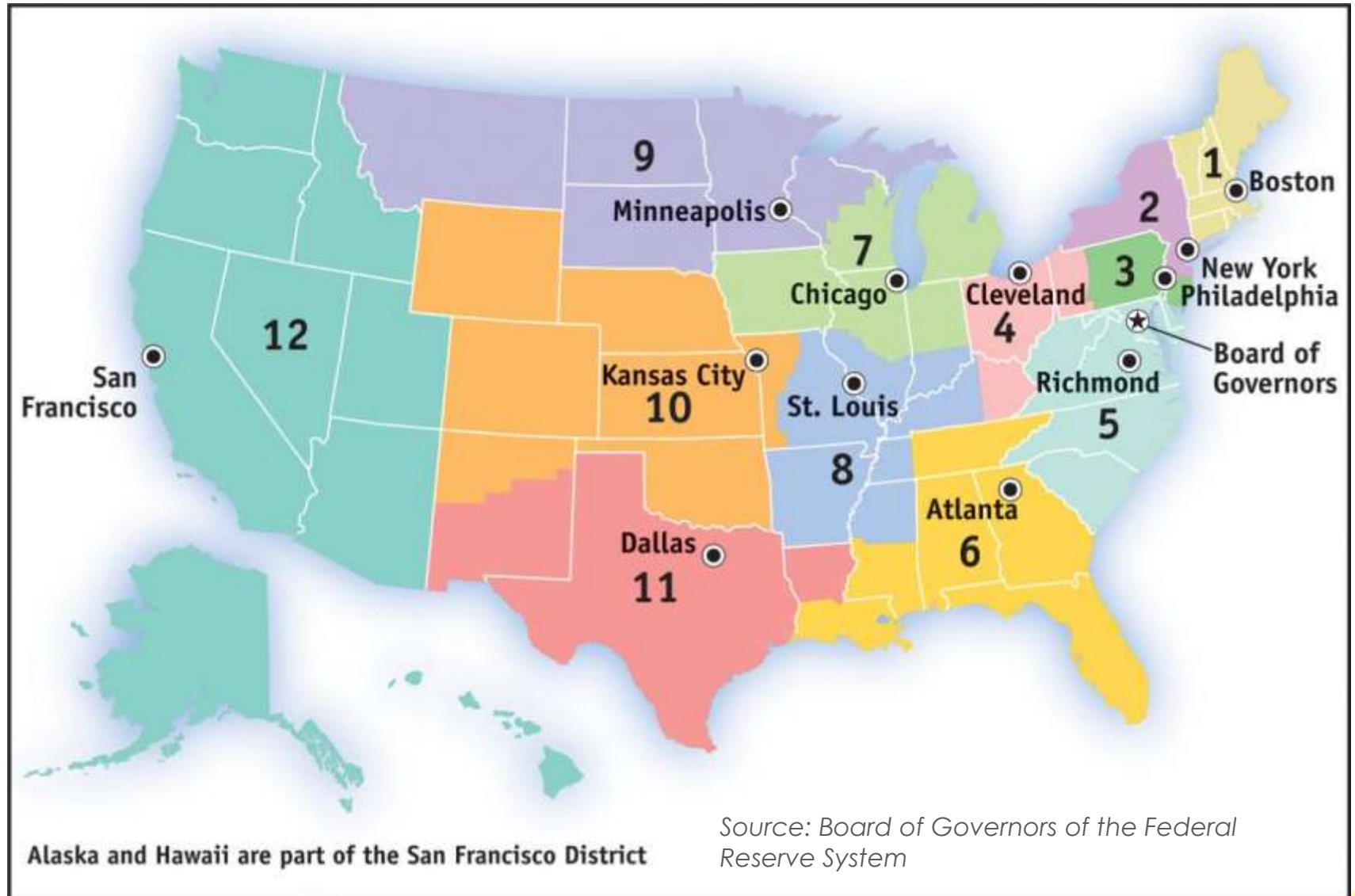
The **Federal Reserve** is a **central bank**; *it oversees and regulates the banking system and controls the monetary base.*

Janet Yellen

15th chair of the Fed
Assumed office
2/3, 2014



THE FEDERAL RESERVE SYSTEM



RESERVE REQUIREMENTS AND THE DISCOUNT RATE

What does a bank do if it can't meet the Fed's reserve requirement?

The ***federal funds market*** allows banks that fall short of the reserve requirement to borrow funds from banks with excess reserves.

The ***federal funds rate*** is the interest rate determined in the federal funds market.

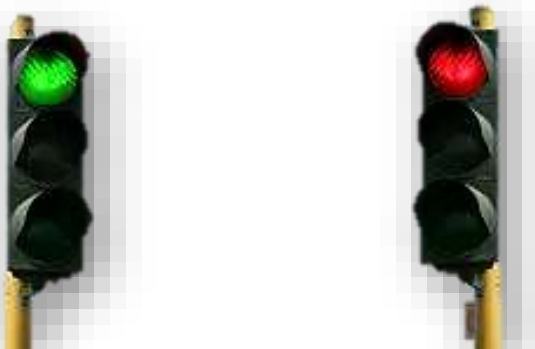


LIBOR (London interbank offered rate) is Europe's equivalent to the federal funds rate.

RESERVE REQUIREMENTS AND THE DISCOUNT RATE

The ***discount rate*** is the rate of interest the Fed charges on loans to banks.

Normally, the discount rate is set 1 percentage point above the federal funds rate in order to discourage banks from turning to the Fed when they are in need of reserves.



COUPON BOND



THE UNITED STATES OF AMERICA

26925E

4 1/4%
TREASURY BOND
OF 1975-85

Dated April 5, 1960
Due May 15, 1985
Redeemable on and after
May 15, 1975

FOR VALUE RECEIVED

PROMISES TO PAY TO THE BEARER THE SUM OF

TEN THOUSAND DOLLARS

ON MAY 15, 1985, AND TO PAY INTEREST ON THE PRINCIPAL SUM FROM THE DATE HEREOF, AT THE RATE OF FOUR AND ONE-QUARTER PERCENT PER ANNUM, PAYABLE ON A SEMI-ANNUAL BASIS ON NOVEMBER 15, 1980, AND THEREAFTER ON MAY 15 AND NOVEMBER 15 IN EACH YEAR UNTIL THE PRINCIPAL HEREOF SHALL BE PAYABLE, UPON PRESENTATION AND SURRENDER OF THE INTEREST COUPONS HERETO ATTACHED AS THEY SEVERALLY MATURE, AT THE TREASURY DEPARTMENT, WASHINGTON, D. C., OR, AT THE HOLDER'S OPTION, AT ANY AGENCY OR AGENCIES IN THE UNITED STATES WHICH THE SECRETARY OF THE TREASURY MAY FROM TIME TO TIME DESIGNATE FOR THE PURPOSE. THIS BOND IS ONE OF A SERIES OF BONDS OF THE UNITED STATES, AUTHORIZED BY THE SECOND LIBERTY BOND ACT, AS AMENDED, ISSUED PURSUANT TO AND SUBJECT TO THE TERMS AND CONDITIONS OF TREASURY DEPARTMENT CIRCULAR NO. 1040, DATED APRIL 4, 1960, AND DESIGNATED 4 1/4 PERCENT TREASURY BONDS OF 1975-85. ALL OR ANY OF THE BONDS OF SAID SERIES MAY BE REDEEMED, AT THE OPTION OF THE UNITED STATES, ON AND AFTER MAY 15, 1975, AT PAR AND ACCRUED INTEREST, ON ANY INTEREST DAY OR DAYS, ON FOUR MONTHS' NOTICE OF REDEMPTION GIVEN IN SUCH MANNER AS THE SECRETARY OF THE TREASURY SHALL PRESCRIBE. IN CASE OF PARTIAL REDEMPTION THE BONDS TO BE REDEEMED WILL BE WITHHOLDEN BY SUCH METHOD AS MAY BE PRESCRIBED BY THE SECRETARY OF THE TREASURY. FROM THE DATE OF REDEMPTION DOCUMENTED IN ANY SUCH NOTICE, INTEREST ON THE BOND CALLED FOR REDEMPTION SHALL CEASE. THE INCOME DERIVED FROM THIS BOND IS SUBJECT TO ALL TAXES IMPOSED UNDER THE INTERNAL REVENUE CODE OF 1954. THIS BOND IS SUBJECT TO ESTATE, INHERITANCE, GIFT OR OTHER ENCISE TAXES, WHETHER FEDERAL OR STATE, BUT IS EXEMPT FROM ALL TAXATION NOW OR HEREAFTER IMPOSED ON THE PRINCIPAL OR INTEREST HEREOF BY ANY STATE, OR ANY OF THE POSSESSIONS OF THE UNITED STATES, OR BY ANY LOCAL TAXING AUTHORITY. THIS BOND, UPON THE DEATH OF THE OWNER, WILL BE REDEEMED AT THE OPTION OF THE DULY CONSTITUTED REPRESENTATIVES OF THE DECEASED OWNER'S ESTATE, AT PAR AND ACCRUED INTEREST, IF IT CONSTITUTES PART OF SUCH ESTATE AND THE PROCEEDS ARE TO BE APPLIED TO THE PAYMENT OF

OPEN-MARKET OPERATIONS

When the Fed buys anything (even apples), reserves increase.

Because government bonds can be stored indefinitely and are easy to buy and sell on the open market, the Fed chooses this commodity to buy and sell daily.

The Fed can buy and sell billions of dollars' worth of government bonds in minutes.

The Fed usually buys and sells short-term bonds called Treasury bills, or T-bills

OPEN-MARKET OPERATIONS

The Fed does not buy bonds directly from the government.

When a central bank buys government debt directly from the government, it is lending directly to the government—in effect printing money to finance the government's budget deficit.

The Fed's decisions are independent of the government.

OPEN-MARKET OPERATIONS

What it looks like:

Open-market operations are the principal tool of monetary policy: the Fed can increase or reduce the monetary base by **buying or selling government debt to banks.**

The Federal Reserve's Assets and Liabilities:

Assets	Liabilities
Government debt (Treasury bills)	Monetary base (currency in circulation + bank reserves)

OPEN-MARKET OPERATIONS

To *increase* the money supply, the Fed must inject reserves into the system.

\$



To *decrease* the money supply, the Fed must remove reserves from the system.

\$



OPEN-MARKET OPERATIONS

An open-market purchase of \$100 million: banks gain reserves.

	Assets		Liabilities	
Federal Reserve	Treasury bills	+\$100 million	Monetary base	+\$100 million

	Assets		Liabilities
Commercial banks	Treasury bills	-\$100 million	No change
	Reserves	+\$100 million	

OPEN-MARKET OPERATIONS

An open-market sale of \$100 million: bank reserves decline

	Assets		Liabilities	
Federal Reserve	Treasury bills	-\$100 million	Monetary base	-\$100 million

	Assets		Liabilities
Commercial banks	Treasury bills	+\$100 million	No change
	Reserves	-\$100 million	

SHOW ME THE MONEY

Q: Where does the Fed get the funds to purchase U.S. Treasury bills from banks?

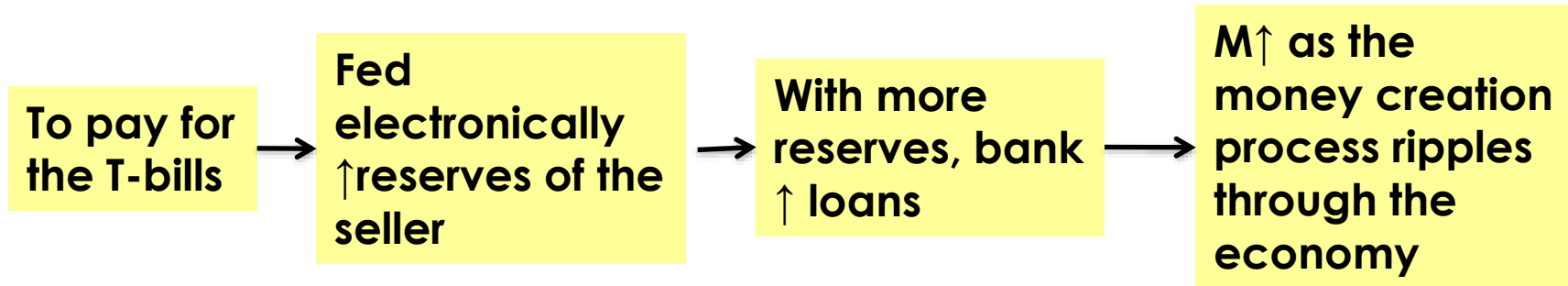
A: It simply creates them with a stroke of the pen (a click of the mouse) that credits the banks' accounts with extra reserves.

(The Fed does NOT need to print money.)

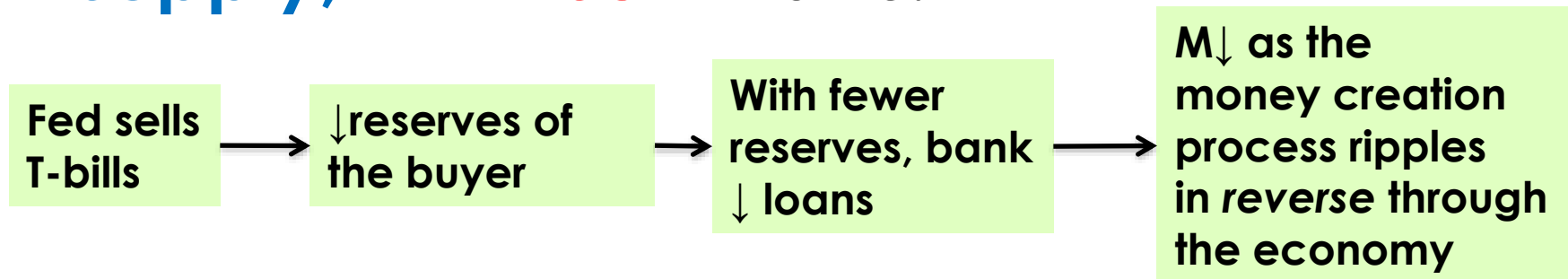


HOW OPEN-MARKET OPERATIONS WORK

If the Fed wants to **increase the money supply**, it will **buy** T-bills:



If the Fed wants to **decrease the money supply**, it will **sell** T-bills:



LEARN BY DOING: PRACTICE QUESTION

Assets		Liabilities	
Required reserves	\$100	Deposits	\$1,000
Loans	\$400		
Treasury bills	\$800		

Suppose this is the only bank and all money is held in this bank. There are no excess reserves. If the Fed makes an open-market sale of \$50 worth of T-bills to this bank, what will happen to the money supply after all adjustments are made?

- a) The money supply will increase by \$50.**
- b) The money supply will decrease by \$50.**
- c) The money supply will increase by \$500.**
- d) The money supply will decrease by \$500.**

THE FINANCIAL CRISIS OF 2008

2008—The worldwide financial system was in crisis, and banks and other financial institutions wanted to borrow more than \$2 trillion. What could be done?



LEARN BY DOING: PRACTICE QUESTION

The Federal Funds rate is the rate that banks are charged when they borrow from the Fed.

- a) True**
- b) False**